

Yassin Hesham

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Career Objective

Passionate Data Scientist and Machine Learning enthusiast dedicated to solving complex real-world problems. Seeking to join an ambitious team where I can continually self-improve, make a meaningful impact in my community, and drive success and growth.

Education

New Giza University, BSc in Information Technology Oct 2021 – July 2025

- **Coursework:** Programming, Data Structures, Data Science, Artificial Intelligence, Database, Machine Learning, Neural and Deep Learning Models, Applied Data Mining, Natural Language Processing, Big Data

Udacity: Intermediate Python Developer Nanodegree May 2023

Experience

AI Engineer / Data Scientist, Studengator – Cairo, Egypt Aug 2025 – Present

- **AI/ML Systems & LLM Engineering:**
 - Migrated vector store to Qdrant, reducing query latency by 40% with improved vector visualization and collection management.
 - Implemented a novel hybrid search method that boosted document retrieval accuracy by 50%.
 - Developed intelligent metadata filter extraction from user queries using LLM-based parsing, dynamically applying filters (school type, location, program duration) to improve RAG retrieval precision.
 - Optimized RAG prompt templates through iterative testing, reducing hallucinations and improving response relevance across the AI assistant.
 - Developed an NLP-based churn classification system categorising student cancellations across 9 behavioural archetypes (backup option, budget overrun, ghosted, visa issue, etc.) with a genuine-intent detection flag, separating real losses from never-serious leads.
- **Machine Learning & Predictive Modelling:**
 - Trained a binary enrollment-prediction model across 5 architectures (Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM) on 3,609 student records, using Optuna (50-trial Bayesian search, 5-fold CV) to tune tree ensembles and CalibratedClassifierCV (isotonic) to produce calibrated win-probability outputs — achieving ROC-AUC 0.674 on a held-out test set of 903 students.
 - Deployed the best model (Random Forest, tuned + calibrated) to score all 4,512 students with win_probability and a percentile-ranked 1–10 lead score exported for sales team prioritisation; feature set (17 features) included target-encoded school and nationality, ordinal academic-level encoding across 14 levels, and log-transformed tuition price.
- **Data Engineering & Analytics Pipelines:**
 - Built a LangGraph-based parallel extraction pipeline that automated collection of 20+ metadata fields (campus locations, Google Maps links, accreditations, rankings) across 145 schools and 1,500+ programs, feeding the Studengator product website.
 - Built an end-to-end student acquisition analytics pipeline over 4,512 applications (2022–2025), engineering a price enrichment layer (fuzzy matching + Gemini Search fallback, 99% coverage) and producing a management report covering funnel performance, win rates by nationality, YoY growth (30% CAGR), and an ideal student profile.
 - Engineered a bot-resistant LinkedIn alumni scraping pipeline (Selenium, 5 layered extraction strategies, MD5 deduplication, incremental checkpointing) targeting 1,200+ alumni profiles per session for sales outreach targeting, packaged as a gated desktop app with real-time logging and a Streamlit dashboard surfacing conversion KPIs to upper management.

Data Scientist / ML Engineer, DSquares – Giza, Egypt

Jul 2023 – Sep 2023

- Built a classification model to predict customer age groups based on behavioural patterns in spending and loyalty activity.
- Conducted end-to-end data analysis, cleaning, preprocessing, and statistical modelling.
- Developed a Recency-Frequency-Monetary (RFM) table and categorised merchants for business insight.

University Student Worker Program, New Giza University – Giza, Egypt

July 2022 – Sep 2022

- Admissions Office: Managed documentation and addressed inquiries from prospective students.
- Marketing Office: Analysed Facebook messaging data and prepared analytical summaries.

Graduation Project

Tailored Learning Platform (TailorEd)

AI-powered E-learning System

- Designed and implemented core AI-driven components: course assistant chatbot, automatic quiz generator, smart semantic content search, and personalised learning path engine.
- Led end-to-end development of a course-specific RAG chatbot using LangChain + OpenAI embeddings with context retrieval from Qdrant for lecture assistance.
- Developed the personalised learning engine: clustered students (A–D) using performance data and education system weighting (IGCSE, American, etc.) to generate tailored study plans.
- Built an automatic quiz maker leveraging LLMs and PDF parsing to generate formative questions from lecture material.
- Engineered the smart content pipeline: chunked, embedded, and summarised lectures using Gemini & OpenAI APIs; implemented chunking strategies and embedding optimisation for semantic search.
- Developed backend APIs (Flask), AI inference pipelines, and integrated MongoDB + Qdrant for student data and vector storage.
- Deployed and maintained the full system on Railway with continuous updates to APIs and LLM prompt templates.
- **Technologies:** Flask, MongoDB, LangChain, OpenAI, Qdrant, Railway, Google Gemini, Scikit-Learn, Transformers

Projects

Lead Prediction Using Nawy Dataset

Python, ML, Real Dataset

- Built an ML pipeline on Nawy's real estate dataset to classify whether a website visitor is likely to convert into a lead.
- End-to-end: preprocessing, feature engineering, model selection, and evaluation.
- Tools Used: Python, Scikit-learn, Pandas

Movie Recommendation Using BERT-RNN

NLP, Python

- Implemented a personalised movie recommender using a multi-task learning architecture.
- Combined BERT embeddings with RNN decoder (based on Stanford NLP paper).
- Tools Used: Python, PyTorch, Transformers

Flight Phase Prediction (Big Data Project)

Big Data, Spark, Hadoop

- Designed a real-time data pipeline using Spark Streaming + OpenSky API to infer flight phases.
- Performed ingestion, preprocessing, feature extraction, and live prediction.
- Tools Used: Spark, Hadoop, Python, OpenSkyAPI, FR24 API

RAG Python Tutor Chatbot

LangChain, Streamlit

- Created an LLM-based assistant that answers coding questions using PDF document retrieval with chunking and

embedding strategies.

- Tools Used: LangChain, OpenAI, Chroma

Premier League Stats Dashboard

Python, MongoDB, Streamlit

- Built a dashboard using Premier League API data and a MongoDB backend.
- Enabled user-friendly queries on player stats, scores, and team data.

Tennis Analysis using YOLOv8

Python, OpenCV

- Built a pipeline to track tennis players and balls using YOLOv8 and court keypoint extraction.

Volunteering Experience

- **Student Union President (School of IT):** Organised tutoring, student initiatives, and collaborative spaces.
- **Team Leader (Graduation/Cap & Gown 2024):** Oversaw logistics and volunteer coordination for 30+ volunteers.
- **Basic Programming Tutor:** Guided first-year students in Python and basic programming logic.
- **TEDxNGU Speaker Hunter Head:** Co-led outreach for prominent guest speakers and event coordination.

Technologies

Languages: Python, C, C++, SQL, NoSQL, JavaScript

AI/ML Frameworks: PyTorch, Scikit-learn, XGBoost, LightGBM, Optuna, Transformers (Hugging Face), OpenCV

LLM & RAG Stack: LangChain, LangGraph, LangSmith, OpenAI API, Google Gemini, Prompt Engineering, RAG Pipelines, Embeddings & Chunking Strategies

Data & Databases: Pandas, NumPy, Matplotlib, Seaborn, MySQL, MongoDB, Qdrant (Vector DB), Firebase, SQLite

Deployment & Tools: Flask, Streamlit, React, Railway, Google Cloud Platform, GitHub, Jupyter Notebook, VS Code